

General Physics I

Problem Solving 3

Problems

Problem 1.

A 3.00-kg particle has a velocity of $(3.00\hat{\mathbf{i}} - 4.00\hat{\mathbf{j}})$ m/s. (a) Find its x and y components of momentum. (b) Find the magnitude and direction of its momentum.

Solution 1.

$$(a) \quad \mathbf{p} = m\mathbf{v} = (9.00\hat{\mathbf{i}} - 12.0\hat{\mathbf{j}}) \text{ kg} \cdot \text{m/s}$$

$$p_x = 9.00 \text{ kg} \cdot \text{m/s}$$

$$p_y = -12.0 \text{ kg} \cdot \text{m/s}$$

$$(b) \quad p = \sqrt{p_x^2 + p_y^2} = \sqrt{(9.00)^2 + (12.0)^2} = 15.0 \text{ kg} \cdot \text{m/s}$$

$$\theta = \tan^{-1}\left(\frac{p_y}{p_x}\right) = \tan^{-1}(-1.33) = 307^\circ$$

Problem 2.

Two blocks of masses m and $3m$ are placed on a frictionless, horizontal surface. A light spring is attached to the more massive block, and the blocks are pushed together with the spring between them (Fig. P9.11). A cord initially holding the blocks together is burned; after that happens, the block of mass $3m$ moves to the right with a speed of 2.00 m/s . (a) What is the velocity of the block of mass m ? (b) Find the system's original elastic potential energy, taking $m = 0.350 \text{ kg}$.

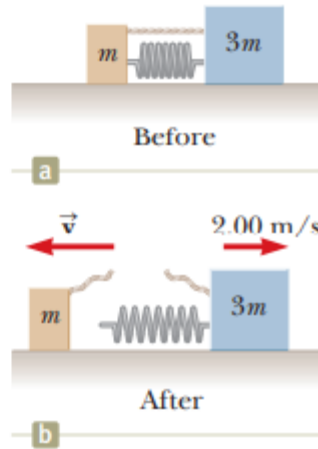


Figure P9.11

Solution 2.

$$(a) \quad p_i = p_f$$

$$0 = Mv_m + (3M)(2.00 \text{ m/s})$$

$$v_m = \boxed{-6.00 \text{ m/s}}$$

$$(b) \quad \frac{1}{2}kx^2 = \frac{1}{2}Mv_M^2 + \frac{1}{2}(3M)v_{3M}^2 = \boxed{8.40 \text{ J}}$$

Problem 3.

An estimated force–time curve for a baseball struck by a bat is shown in Figure P9.13. From this curve, determine (a) the magnitude of the impulse delivered to the ball and (b) the average force exerted on the ball.

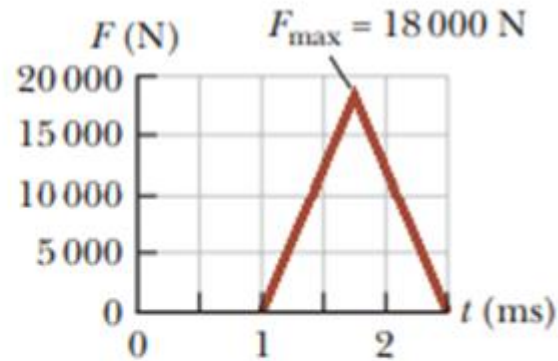


Figure P9.13

Solution 3.

Impulse = The area under the curve

$$(a) \quad I = \frac{1}{2} (1.50 \times 10^{-3} \text{ s})(18\,000 \text{ N}) = \boxed{13.5 \text{ N} \cdot \text{s}}$$

$$(b) \quad F = \frac{13.5 \text{ N} \cdot \text{s}}{1.50 \times 10^{-3} \text{ s}} = \boxed{9.00 \text{ kN}}$$

$$(c) \quad F_{\max} = \boxed{18.0 \text{ kN}}$$

Problem 4.

A block of mass $m = 5.00 \text{ kg}$ is released from point ① and slides on the frictionless track shown in Figure P8.6. Determine (a) the block's speed at points ② and ③ and (b) the net work done by the gravitational force on the block as it moves from point ① to point ③.

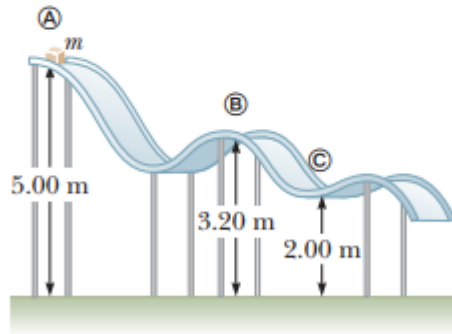


Figure P8.6

Solution 4.

$$(a) \quad (\Delta K)_{A \rightarrow B} = \Sigma W = W_g = mg\Delta h = mg(5.00 - 3.20)$$

$$\frac{1}{2} mv_B^2 - \frac{1}{2} mv_A^2 = m(9.80)(1.80)$$

$$v_B = \boxed{5.94 \text{ m/s}}$$

$$v_C = \sqrt{v_A^2 + 2g(5.00 - 2.00)} = \boxed{7.67 \text{ m/s}}$$

$$(b) \quad W_{g \& A \rightarrow C} = mg(3.00 \text{ m}) = \boxed{147 \text{ J}}$$

Problem 5.

Two objects are connected by a light string passing over a light, frictionless pulley as shown in Figure P8.7. The object of mass $m_1 = 5.00$ kg is released from rest at a height $h = 4.00$ m above the table. Using the isolated system model, (a) determine the speed of the object of mass $m_2 = 3.00$ kg just as the 5.00-kg object hits the table and (b) find the maximum height above the table to which the 3.00-kg object rises.

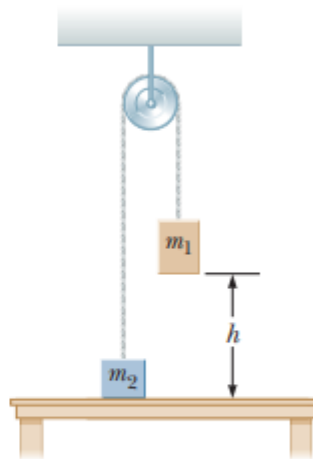


Figure P8.7

Solution 5.

$$(a) \quad (5.00 \text{ kg})g(4.00 \text{ m}) = (3.00 \text{ kg})g(4.00 \text{ m}) + \frac{1}{2}(5.00 + 3.00) v^2$$

$$v = \sqrt{19.6} = \boxed{4.43 \text{ m/s}}$$

$$(b) \quad \frac{1}{2}(3.00) v^2 = mg \Delta y = 3.00g \Delta y$$

$$\Delta y = 1.00 \text{ m}$$

$$y_{\text{max}} = 4.00 \text{ m} + \Delta y = \boxed{5.00 \text{ m}}$$

Problem 6.

A 5.00-kg block is set into motion up an inclined plane with an initial speed of $v_i = 8.00 \text{ m/s}$ (Fig. P8.23). The block comes to rest after traveling $d = 3.00 \text{ m}$ along the plane, which is inclined at an angle of $\theta = 30.0^\circ$ to the horizontal. For this motion, determine (a) the change in the block's kinetic energy, (b) the change in the potential energy of the block-Earth system, and (c) the friction force exerted on the block (assumed to be constant). (d) What is the coefficient of kinetic friction?

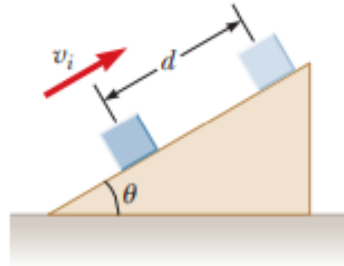


Figure P8.23

Solution 6.

$$(a) \quad \Delta K = \frac{1}{2} m(v^2 - v_i^2) = -\frac{1}{2} mv_i^2 = \boxed{-160 \text{ J}}$$

$$(b) \quad \Delta U = mg(3.00 \text{ m}) \sin 30.0^\circ = \boxed{73.5 \text{ J}}$$

$$(c) \quad \text{The energy lost to friction is } 86.5 \text{ J}$$

$$f = \frac{86.5 \text{ J}}{3.00 \text{ m}} = \boxed{28.8 \text{ N}}$$

$$(d) \quad f = \mu_k n = \mu_k mg \cos 30.0^\circ = 28.8 \text{ N}$$

$$\mu = \frac{28.8 \text{ N}}{(5.00 \text{ kg})(9.80 \text{ m/s}^2) \cos 30.0^\circ} = \boxed{0.679}$$

Problem 7.

A block of mass $m = 2.00$ kg is attached to a spring of force constant $k = 500$ N/m as shown in Figure P8.15. The block is pulled to a position $x_i = 5.00$ cm to the right of equilibrium and released from rest. Find the speed the block has as it passes through equilibrium if (a) the horizontal surface is frictionless and (b) the coefficient of friction between block and surface is $\mu_k = 0.350$.

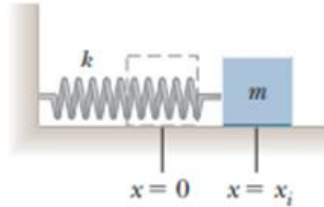


Figure P8.15

Solution 7.

$$(a) \quad W_s = \frac{1}{2} kx_i^2 - \frac{1}{2} kx_f^2 = \frac{1}{2} (500) (5.00 \times 10^{-2})^2 - 0$$

$$= 0.625 \text{ J}$$

$$W_s = \frac{1}{2} mv_f^2 - \frac{1}{2} mv_i^2 = \frac{1}{2} mv_f^2 - 0$$

$$v_f = \sqrt{\frac{2(\sum W)}{m}} = \sqrt{\frac{2(0.625)}{2.00}} \text{ m/s} = \boxed{0.791 \text{ m/s}}$$

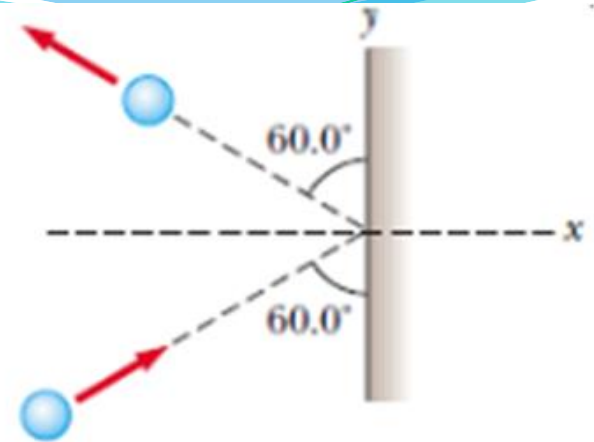
$$(b) \quad \frac{1}{2} mv_i^2 - f_k \Delta x + W_s = \frac{1}{2} mv_f^2$$

$$0 - (0.350)(2.00)(9.80)(0.0500) \text{ J} + 0.625 \text{ J} = \frac{1}{2} mv_f^2$$

$$0.282 \text{ J} = \frac{1}{2} (2.00 \text{ kg}) v_f^2$$

$$v_f = \sqrt{\frac{2(0.282)}{2.00}} \text{ m/s} = \boxed{0.531 \text{ m/s}}$$

Problem 8. A 3 kg steel ball strikes a wall with a speed of 10 m/s at an angle of 60° with the surface. It bounces off with the same speed and angle. If the ball contact with the wall for 0.2 s, what is the average force exerted by the wall on the ball?



Solution 8.

$$\Delta \mathbf{p} = \mathbf{F} \Delta t$$

$$\Delta p_y = m(v_{fy} - v_{iy}) = m(v \cos 60.0^\circ) - mv \cos 60.0^\circ = 0$$

$$\Delta p_x = m(-v \sin 60.0^\circ - v \sin 60.0^\circ) = -2mv \sin 60.0^\circ$$

$$= -2(3.00 \text{ kg})(10.0 \text{ m/s})(0.866)$$

$$= -52.0 \text{ kg} \cdot \text{m/s}$$

$$F_{\text{ave}} = \frac{\Delta p_x}{\Delta t} = \frac{-52.0 \text{ kg} \cdot \text{m/s}}{0.200 \text{ s}} = \boxed{-260 \text{ N}}$$

REFERENCES

